Pixels Without Geometry? Auditing the 3D Consistency of Video World Models with Structure-from-Motion

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Abstract: Geometric scores such as SfM registration or reprojection error are increasingly used to judge whether video models can serve as robot world models. We show that these scores cannot be read without motion-aware controls. On real DROID wrist-camera clips, three image-to-video models (LTX-Video, CogVideoX, Wan2.2) and an action-conditioned DROID world model (Ctrl-World) predict the next five seconds. We score the predictions with COLMAP, epipolar geometry and VGGT, and compare each clip against controls built from real footage: non-rigid warps, sharpness-matched blur, unrelated frames, a frozen first frame and, per clip, same-episode real footage matched to the clip's image motion. Raw comparisons with the real continuation are misleading: a frozen frame beats real footage on match failure, inliers and VGGT depth disagreement. Against motion-matched references, the apparent advantages of two near-static generators shrink or vanish, and paired tests show that matching significantly shifts several comparisons. As a secondary, sample-sensitive observation, Ctrl-World shows no detectable difference from the real continuation in VGGT disagreement but is worse than motion-matched footage on the shared clips, not significantly so on all 60. We distill the controls into a reporting protocol for geometric evaluation of robot video predictions.

Keywords: World models, video prediction, structure-from-motion, evaluation

1 Introduction

Whether robots need world models is usually argued through downstream task success. A prior question is whether a learned model's predicted future of the robot's camera is *geometrically consistent*: one rigid scene, seen by a camera that moves as the robot moves. Structure-from-motion (SfM) and feed-forward reconstruction offer an operational test, and they are increasingly used to score generated video. Raw geometric scores are easy to misread, however. A frozen camera yields clean matches and low errors without informing about 3D structure, blur removes features without changing geometry, and non-rigid distortion can still register.

We therefore calibrate every score against controls built from the real continuation, and evaluate general image-to-video models and an action-conditioned robot world model on real DROID wrist-camera clips. Contributions: (i) a control ladder (non-rigid warps, sharpness-matched blur, unrelated frames, a frozen frame) that shows what COLMAP, epipolar and VGGT scores respond to; (ii) per-clip, same-episode motion-matched real references, which show with paired confidence intervals that raw comparisons flatter near-static predictions and can hide deficits in moving ones; and (iii) a repeatable reporting protocol: report image motion and match coverage, compare against matched real references, and keep the scores separate rather than combining them.

2 Related Work

Classical geometry has been applied to generated video before: Li et al. [1] measured SIFT/RANSAC epipolar consistency and COLMAP reconstructions on a handful of Sora, Gen-2 and Pika clips, and MEt3R [2], WorldScore [3] and GeoT2V-Bench [4] score 3D consistency with feed-forward reconstructors [5, 6, 7]. Camera-controllable generators report pose error and COLMAP success rates [8, 9, 10], and broader benchmarks probe plausibility [11, 12, 13]. Several of the confounds we study have been observed before. MEt3R [2] notes that shorter camera trajectories score as more consistent and that blur-sensitive metrics can favor generated over real footage. GeoT2V-Bench [4] reports image motion and disagreements between geometry measures. RoboPhys-3D [14] finds generated robot videos matching or exceeding ground truth on geometric consistency despite wrong scene state. What these works do not provide is a controlled perturbation study on real footage that quantifies how each confound changes model comparisons. For robots, video models serve as simulators and planners [15, 16, 17, 18, 19, 20], and action-conditioned models trained on real data, such as Ctrl-World [21] on DROID [22], target policy evaluation in imagination. Robot world models are mostly evaluated by policy-ranking agreement or visual fidelity [23, 24, 25]. Closest to us, RoboPhys-3D [14] reconstructs generated robot videos with VGGT, COLMAP and SLAM, but in simulation (RoboTwin) and without perturbed real references. We do not claim the first SfM analysis of generated video, nor the first observation of these confounds. We contribute paired, per-clip controls on real robot wrist-camera footage that show how motion and appearance change geometric model comparisons.

3 Method

Clips and models. From the DROID-100 sample [22] (320×180, 15 fps) we take, per episode, the 5 s wrist-camera window with the most image motion. A fixed mask removes the gripper, which never moves in the wrist view. LTX-Video [26], CogVideoX-5B [27] and Wan2.2-5B [28] receive the first frame and the episode instruction. Ctrl-World [21] receives the first frame of all three DROID cameras and the *recorded* end-effector poses. It rolls out at 5 Hz in overlapping five-frame chunks, each advancing four frames (0.8 s). Ctrl-World thus differs from the other models in its inputs, architecture and training data, not only in action conditioning. Outputs are resampled to 320×180 and share the real first frame. Generation ran in episode order and never failed. Compute (one 16 GB laptop GPU) limited the slowest model, Wan2.2, to the first 25 clips, so these 25 are the clips shared by all models. LTX and Ctrl-World cover all 60 (Appendix A).

Controls. All controls are real footage and share the generators' H.264 round trip: *real*, the true continuation; *warp-*a, an independent smooth non-rigid warp per frame ($a \in \{1, 2, 4, 8\}$ px); *blurred*, Gaussian blur matched to the softest outputs' sharpness; *slowed*, the first 40% stretched over 5 s; *unrelated*, frames from other clips; and *frozen*, the first frame repeated. *Motion-matched* references are chosen per generated clip from real footage of the same episode (same scene and camera): the true continuation slowed by up to 3× (no repeated frames at 5 fps) or any other 5 s window of the episode. We keep the candidate closest in image motion if it is within 25%, and repeat the matching with two independent motion estimators (below).

Metrics. All metrics use the 5 s horizon at 5 fps (every third frame, so Ctrl-World's 5 Hz output has no repeated frames). We use COLMAP [29] registration rate (sequential matching with an overlap of 10 frames) and pairwise metrics on frame pairs 0.4 s apart: SIFT [30] + MAGSAC++ [31] epipolar inliers, Sampson error and match-failure rate. Pairs with too few matches count as failures, and Sampson error averages only pairs where matching and fundamental-matrix estimation succeed. Median SIFT-match displacement serves as an *image-motion proxy*. It reflects camera motion, but also scene depth, moving objects and which features survive matching, so we also match with median dense Farneback optical flow, which does not depend on matched features. VGGT [7] depth disagreement is the median relative difference when frame i's confident depth is reprojected into frame j. It measures the internal consistency of VGGT's own depth and pose predictions, not ground-truth geometry. We therefore report the fraction of points that land in frame (coverage) and

	Motion	Reg.	Fail	Inliers	Sampson	VGGT	Cov.	Path
Real	24.2 ± 1.6	0.59 ± 0.08	0.11 ± 0.04	53.7 ± 8.5	0.32 ± 0.02	0.028 ± 0.003	0.87 ± 0.01	0.017 ± 0.001
Slowed 2.5×	10.9 ± 0.9	0.66 ± 0.07	0.060 ± 0.041	83.8 ± 11.2	0.26 ± 0.02	0.013 ± 0.001	0.92 ± 0.01	0.11 ± 0.01
Blurred	23.7 ± 1.5	0.76 ± 0.07	0.15 ± 0.04	57.4 ± 9.6	0.30 ± 0.02	0.029 ± 0.003	0.87 ± 0.01	0.028 ± 0.003
Warp 8 px	24.6 ± 1.3	0.70 ± 0.07	0.15 ± 0.04	43.0 ± 6.4	0.49 ± 0.03	0.045 ± 0.005	0.85 ± 0.01	0.047 ± 0.004
Unrelated	118.3 ± 6.4	0.000 ± 0.000	0.84 ± 0.02	14.8 ± 0.8	0.24 ± 0.10	0.67 ± 0.08	0.18 ± 0.01	0.17 ± 0.01
LTX	10.3 ± 2.0	0.25 ± 0.06	0.32 ± 0.07	38.8 ± 6.1	0.24 ± 0.03	0.024 ± 0.005	0.92 ± 0.02	0.15 ± 0.01
CogVideoX	2.40 ± 0.46	0.20 ± 0.05	0.023 ± 0.020	132.7 ± 22.3	0.14 ± 0.01	0.007 ± 0.001	0.98 ± 0.01	0.12 ± 0.01
Wan2.2	2.57 ± 0.64	0.27 ± 0.07	0.043 ± 0.028	122.7 ± 17.6	0.16 ± 0.02	0.011 ± 0.002	0.99 ± 0.00	0.13 ± 0.01
Ctrl-World	19.7 ± 2.1	0.26 ± 0.06	0.36 ± 0.06	24.4 ± 2.0	0.35 ± 0.02	0.031 ± 0.003	0.86 ± 0.02	0.10 ± 0.01

Table 1: 5 s horizon, 25 clips common to all conditions, mean $\pm$ s.e.m. Motion: image-motion proxy (px). Reg.: COLMAP registration rate. Fail: share of 0.4 s frame pairs that fail to match. Sampson in px. VGGT: depth disagreement. Cov.: fraction of reprojected points in frame. Path: camera-path error vs. the VGGT-estimated reference (references: frozen camera 0.175, real path of a different clip 0.115). Valid clips per metric are 22–25.

Model	Reference	Fail	Inliers	VGGT	VGGT (filt.)	Reg.	n
LTX	real	+0.32*	-35.1*	-0.000	-0.001	-0.41*	15
	matched	+0.38*	-53.6*	+0.008	+0.004	-0.32*	15
CogVideoX	real	-0.26*	+16.5	-0.039*	-0.025*	-0.29	6
	matched	-0.083*	+12.2	-0.011*	-0.010*	-0.10	6
Wan2.2	real	-0.18*	+23.8	-0.010	-0.003	-0.20	6
	matched	-0.083	+12.8	+0.005	+0.003	-0.13	6
Ctrl-World	real	+0.23*	-31.4*	+0.000	+0.001	-0.39*	22
	matched	+0.28*	-36.2*	+0.008*	+0.006*	-0.34*	22
Frozen frame	real	-0.073*	+177.3*	-0.031*	-0.023*	-0.64*	25

Table 2: Paired differences, model − reference, on the clips for which a motion-matched reference exists (n; SIFT-displacement matching, median matching error 1–4%). "real": the true continuation. "matched": same-episode real footage matched to the clip's image motion. Pairwise metrics at 0.4 s. *: 95% bootstrap CI excludes zero. Lower is better for Fail and VGGT, higher for Inliers and Reg. Other gaps and dense-flow matching are in Table 4.

a visibility-filtered variant: confident target depth only, with points more than 10% behind the target surface treated as occluded. Camera-path error is the Sim(3)-aligned RMSE between the VGGT camera centers of the prediction and of the real clip, divided by the reference path length. It measures agreement with a *VGGT-estimated reference trajectory* up to scale, not metric ground-truth accuracy. Model-minus-control differences use paired bootstrap 95% CIs (10,000 resamples).

4 Results

What the scores respond to. On the warp ladder (Table 3, Appendix B), reprojection error, Sampson error and default VGGT disagreement rise monotonically with distortion. Filtered VGGT stays flat up to 2 px, then rises. Registration (0.59, 0.73, 0.64, 0.56, 0.70) does not decrease monotonically with distortion. Slowing and blurring do not lower registration (66% and 76% vs. 59%, Table 1), but slowing lowers VGGT disagreement, so image motion shifts the scores.

Raw comparisons reward uninformative output. A frozen first frame beats the real continuation on match failure, inliers and both VGGT variants, all significantly (Table 2), while registering 64 points fewer frames. CogVideoX's and Wan2.2's image motion is about a tenth of the real level (2.4–2.6 vs. 24 px, Table 1). On their matchable clips they too look better than the real continuation (Table 2), and their predicted futures are nearly static views in which objects change (Figure 1).

Motion-matched references shrink or remove apparent advantages. Against same-episode real footage with the same image motion, CogVideoX's VGGT advantage shrinks by about 70% (-0.039 to -0.011) but remains significant, and Wan2.2's disappears (-0.010 to +0.005). The same holds

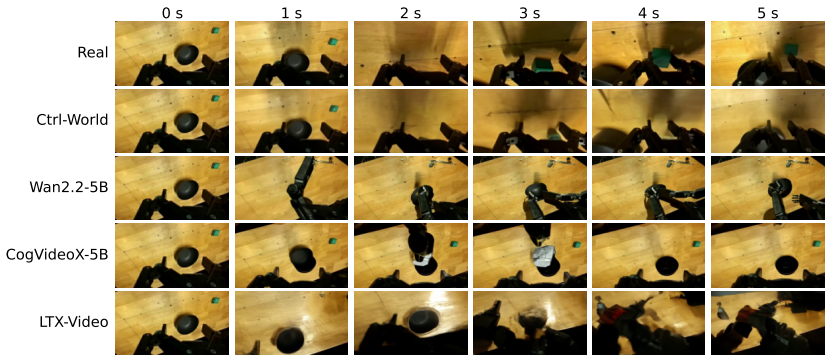


Figure 1: One clip, 0–5 s. Ctrl-World follows the real camera motion (softer, losing texture). Wan2.2 cuts to a third-person view, CogVideoX holds the view while objects morph, and LTX distorts the scene. A median-performance example is in Appendix A.

at the other gaps and with dense-flow matching (Table 4). LTX's significant VGGT advantage at 0.8 s (-0.019, dense matching) vanishes against matched footage, and its match-failure gap widens (+0.32 to +0.38). The shift, raw minus matched (equal to matched reference minus real continuation), is significant for CogVideoX (-0.028 [-0.049, -0.006], SIFT), Ctrl-World (-0.008 [-0.014, -0.003], SIFT) and LTX (-0.012 [-0.023, -0.003], dense), but not for Wan2.2. Only 6–9 of 25 clips could be matched for the near-static models, so per-model estimates are exploratory (intervals in Appendix C). Most matched references are other windows of the episode rather than slowed continuations, and these can differ in viewpoint, objects and occlusion, so the shifts are not attributable to motion alone. Low Sampson error must be read alongside match failures, because it summarizes only pairs where geometric estimation succeeds.

Motion-matched references can expose a hidden deficit. The action-conditioned Ctrl-World model moves the image almost as much as reality (19.7 vs. 24.2 px). Against the real continuation, its VGGT disagreement shows no detectable difference ($+0.000$). Against motion-matched footage on the shared clips it is higher at every gap (+0.007 to +0.018; significant under SIFT matching, mostly not under dense matching), but on all 60 clips the difference is small and not significant (+0.002 to +0.004, Appendix C), so we treat it as secondary and sample-sensitive. Registration is 33 points lower than real and 50 points lower than sharpness-matched blurred real footage. Match failure is 24 and 21 points higher, respectively, so blur at this level does not explain the deficit. With no obvious concentration at rollout-step transitions (Appendix A), this is consistent with temporally unstable texture. No model's VGGT-estimated camera path is reliably closer to the reference than an unrelated clip's real path (Ctrl-World -0.015, CI [-0.034, 0.003]); the path metric separates only near-exact trajectories (Table 1).

5 Discussion and Limitations

Geometric scores of generated video conflate scene consistency, image motion and feature persistence, and comparing against the real continuation alone does not separate them. Per-clip motion-matched references change the conclusions in both directions: they shrink or remove the advantages of near-static predictions and can reveal deficits that the raw comparison hides. We propose a simple protocol: report image motion (with two estimators) and match coverage, compare each clip against a frozen frame and against motion- and appearance-matched real references, and report the scores separately rather than as one number. We do not claim that action conditioning causes Ctrl-World's behavior, because it also differs in inputs, architecture and training data. Nor do we claim that these scores predict planning success. **Limitations:** one dataset and camera at low resolution; 25 shared clips, few matchable for near-static models; small open models; DROID-100 likely in Ctrl-World's training data; arm and object motion break rigidity; VGGT scores reflect its own predictions.

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A Implementation details

Checkpoints (Hugging Face repository at commit). LTX-Video: `Lightricks/LTX-Video`, the 2B diffusers-format weights in the repository root, `8984fa2`. CogVideoX: `zai-org/CogVideoX-5b-I2V`, `a6f0f48`. Wan2.2: `Wan-AI/Wan2.2-TI2V-5B-Diffusers`, `b8fff73`. Ctrl-World: `yjguo/Ctrl-World` (`checkpoint-10000.pt`), `8cf8146`, with `stabilityai/stable-video-diffusion-img2vid` (`9cf024d`) and `openai/clip-vit-base-patch32`. VGGT: `facebook/VGGT-1B`, `860abec`. LTX uses the bf16 T5-v1.1-XXL text encoder from the CogVideoX repository.

Generation. LTX-Video: 704×480, 121 frames at 24 fps, 40 steps. CogVideoX-5B-I2V: 720×480, 49 frames at 8 fps, 50 steps. Wan2.2-TI2V-5B: 1280×704, 121 frames at 24 fps, 40 steps. Ctrl-World: replay of recorded 7-D end-effector states (xyz, roll-pitch-yaw, gripper), 50 steps. It runs seven rollout steps at 5 Hz, each generating five frames that include the previous step's last frame, so each step advances four frames (0.8 s). Of the 29 frames, the first 26 (5 s) are kept. The 5B transformers store weights in fp8 and compute in bf16. One prompt template uses the episode instruction (a neutral instruction where DROID-100 has none). Clips are the 60 highest-motion windows of DROID-100. LTX and Ctrl-World cover all 60, CogVideoX 40 and Wan2.2 25. Main comparisons use the 25 shared clips. Robustness analyses additionally use all 60 available clips.

Scoring. COLMAP incremental SfM with one shared SIMPLE_RADIAL camera, sequential matching (overlap 10), SIFT at doubled resolution, and initialization and registration thresholds lowered to 40 and 15 inliers for 320×180 frames. Epipolar pairs need at least 16 ratio-test matches, or they count as match failures. VGGT runs on the 26 frames of the 5 fps track, and source points are those above the median depth confidence. The slowed control (2.5×) approximately matches LTX's average image motion. The blur ($\sigma = 0.7$ px) matches the Laplacian-variance sharpness of Ctrl-World and LTX. The path-error reference "unrelated clip" averages each clip's error against the real paths of all 24 other clips. The frozen-camera reference collapses all predicted centers to one point.

Chunk transitions. On the 5 fps track, the share of consecutive Ctrl-World frame pairs that fail to match is 16%, 20%, 21% and 23% at the four positions within a rollout step (the last position crosses into the next step), against 3–5% for real footage. We see no obvious concentration at the transition, although this coarse test cannot rule out subtler boundary effects.

Additional example. Figure 2 shows a clip near the median of both Ctrl-World's and LTX's registration rates.

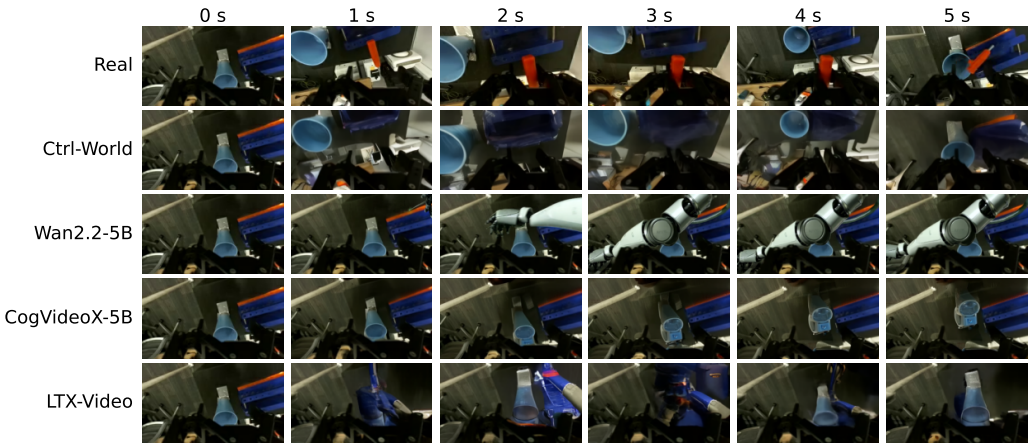


Figure 2: A representative clip (near the median registration of Ctrl-World and LTX). Ctrl-World follows the real camera's sweep. Wan2.2 and CogVideoX hold the viewpoint while objects change, and LTX distorts the scene.

B Calibration ladder

	Real	Warp-1	Warp-2	Warp-4	Warp-8	Unrelated
Reprojection error (px) ↓	0.48	0.60	0.61	0.85	1.08	–
Median Sampson error (px) ↓	0.32	0.33	0.35	0.39	0.49	0.24
Epipolar inliers per pair ↑	53.7	54.6	55.1	50.3	43.0	14.8
Pairs failing to match ↓	0.11	0.11	0.14	0.10	0.15	0.84
VGGT depth disagreement ↓	0.028	0.029	0.030	0.032	0.045	0.67
VGGT, visibility-filtered ↓	0.022	0.022	0.022	0.024	0.030	0.20
Registration rate	0.59	0.73	0.64	0.56	0.70	0.000

Table 3: Calibration ladder at the main settings (5 fps, 0.4 s pairs, the 25 shared clips). "–": too few frames register.

C Robustness of the motion-matched comparison

Model	Matching	0.2 s	0.4 s	0.8 s
LTX	SIFT (n=15)	+0.002 → +0.007	-0.000 → +0.008	-0.017 → +0.000
	dense flow (n=18)	+0.001 → +0.008	-0.003 → +0.010	-0.019* → +0.006
CogVideoX	SIFT (n=6)	-0.019* → -0.010*	-0.039* → -0.011*	-0.065* → -0.013*
	dense flow (n=9)	-0.016* → -0.014*	-0.031* → -0.022*	-0.051* → -0.033*
Wan2.2	SIFT (n=6)	-0.006 → -0.000	-0.010 → +0.005	-0.029 → +0.011*
	dense flow (n=7)	-0.009 → -0.002	-0.015 → +0.001	-0.036* → -0.000
Ctrl-World	SIFT (n=22)	+0.003 → +0.007*	+0.000 → +0.008*	+0.001 → +0.018*
	dense flow (n=23)	+0.002 → +0.005*	-0.002 → +0.003	-0.008 → +0.008

Table 4: VGGT depth disagreement, model − real continuation → model − motion-matched real footage, on the same matched clips, at three frame-pair gaps and with two matching estimators (25 shared clips). *: 95% bootstrap CI excludes zero.

Matching. Median relative matching error on matched clips is 1–4% for every model and estimator. Matches exist for 15–18 of 25 LTX clips, 22–23 Ctrl-World clips, and 6–9 CogVideoX and Wan2.2 clips: real wrist footage rarely moves as little as the near-static models' outputs. Matched references are mostly other windows of the same episode rather than slowed continuations (SIFT matching: LTX 3 slowed + 12 windows, CogVideoX 2 + 4, Wan2.2 0 + 6, Ctrl-World 6 + 16). Other windows share scene and camera but can differ in viewpoint, object configuration, occlusion and texture. Table 5 gives complete intervals and a slowed-only analysis where feasible. With 3–6 slowed controls these are exploratory. Where available, the slowed-only shifts have the same sign as the full analysis.

All 60 clips (LTX, Ctrl-World). At 0.4 s with SIFT matching, LTX (39 matched clips) has VGGT differences of +0.003 (raw) and +0.005 (matched), and match-failure differences of +0.24 and +0.27. Ctrl-World (52 matched clips) has VGGT differences of +0.002 and +0.002 (dense matching: +0.000 and +0.004), and match-failure differences of +0.21 and +0.23. The VGGT CIs include zero, while every match-failure CI excludes zero. The direction of the shared-clip results holds, but the VGGT deficit of Ctrl-World against matched footage is not significant on the larger set.

Code. All code will be released after review.

Model	Matching	Controls	VGGT: raw	VGGT: matched	VGGT: shift
LTX	SIFT	all (3+12)	-0.000 [-0.015, +0.013]	+0.008 [-0.006, +0.022]	-0.008 [-0.021, +0.004]
		slowed only (3)	+0.009 [-0.022, +0.053]	+0.028 [-0.001, +0.080]	-0.019 [-0.027, -0.011]
	dense	all (3+15)	-0.003 [-0.016, +0.010]	+0.010 [-0.003, +0.023]	-0.012 [-0.023, -0.003]
		slowed only (3)	-0.003 [-0.006, +0.002]	+0.002 [-0.004, +0.009]	-0.004 [-0.015, +0.002]
CogVideoX	SIFT	all (2+4)	-0.039 [-0.064, -0.015]	-0.011 [-0.020, -0.004]	-0.028 [-0.049, -0.006]
		slowed only (2)	–	–	–
	dense	all (1+8)	-0.031 [-0.051, -0.013]	-0.022 [-0.040, -0.009]	-0.009 [-0.032, +0.012]
		slowed only (1)	–	–	–
Wan2.2	SIFT	all (0+6)	-0.010 [-0.026, +0.006]	+0.005 [-0.004, +0.015]	-0.015 [-0.040, +0.010]
		slowed only (0)	–	–	–
	dense	all (0+7)	-0.015 [-0.031, +0.001]	+0.001 [-0.011, +0.016]	-0.016 [-0.043, +0.010]
		slowed only (0)	–	–	–
Ctrl-World	SIFT	all (6+16)	+0.000 [-0.008, +0.008]	+0.008 [+0.003, +0.014]	-0.008 [-0.014, -0.003]
		slowed only (6)	+0.012 [-0.000, +0.027]	+0.016 [+0.006, +0.029]	-0.004 [-0.010, -0.000]
	dense	all (6+17)	-0.002 [-0.010, +0.006]	+0.003 [-0.004, +0.010]	-0.005 [-0.014, +0.005]
		slowed only (6)	-0.010 [-0.022, +0.002]	-0.000 [-0.015, +0.009]	-0.010 [-0.020, -0.000]

Table 5: VGGT depth disagreement at 0.4 s on the 25 shared clips, mean [95% bootstrap CI]. Raw: model − real continuation. Matched: model − motion-matched reference. Shift: matched reference − real continuation, which equals raw − matched. A negative shift means the model looks worse against the motion-matched reference than against the real continuation. Controls: number of slowed continuations + other episode windows. "–": fewer than three clips.